Radix-2 Booth halves signed multiplication cycles

Why fewer cycles? The supplied baseline processes a 64-bit sign-extended multiplier. Booth recoding handles the signed 32-bit multiplier in 32 iterations.

Baseline implementation	Enhanced implementation
Sign-extends operands to 64 bits	Directly handles signed operands using Booth recoding
Performs 64 shift-and-add iterations	Performs 32 add/subtract-and-shift iterations
Processes the extended multiplier one bit per iteration	Processes one original multiplier bit per iteration

Result: 50% fewer computation cycles for 32-bit signed multiplication.

Unsigned multiplication remains at 32 cycles. No * operator is used in the synthesizable RTL.

How radix-2 Booth updates the hardware

Register	Width for 32-bit operands	Purpose
A	33 bits	Partial-product accumulator
M	33 bits	Sign-extended multiplicand
Q	32 bits	Multiplier, then lower product bits
Qminus1	1 bit	Previous least-significant multiplier bit

{Q[0], Qminus1}	Operation
00, 11	Keep A unchanged
01	A ← A + M
10	A ← A − M

1. Inspect {Q[0], Qminus1} → 2. Add, subtract or hold → 3. Arithmetic right shift → 4. Repeat 32 times

After 32 iterations, the 64-bit product is {A[31:0], Q}.

Why 33-bit A and M? The extra bit supports intermediate arithmetic involving the most negative signed operand.

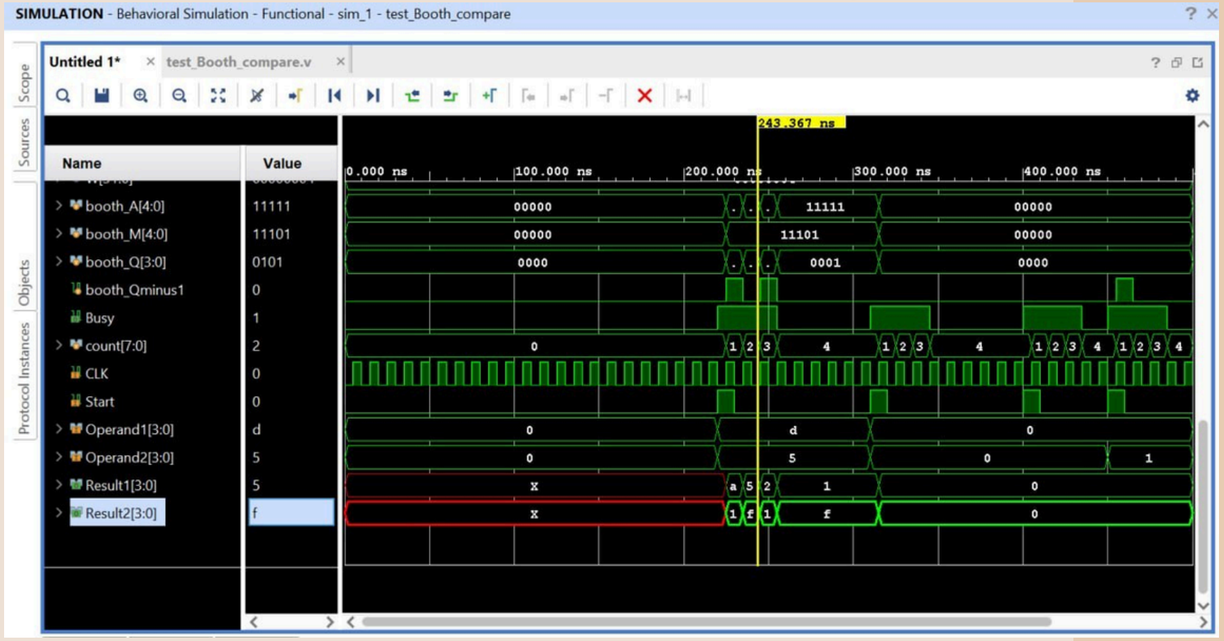
Verified against expected products and the baseline

Worked example: −3 × 5 = −15, using 4-bit operands

Final product = {A[3:0], Q}

= 11110001 = 0xF1 = −15

Iteration	Operation before shift	A after shift	Q after shift	Qminus1
1	A − M	00001	1010	1
2	A + M	11111	0101	0
3	A − M	00001	0010	1
4	A + M	11111	0001	0



Verification	Result
4-bit comparison	514 checks passed, including every signed operand pair
32-bit behavioural	467 checks passed per design
32-bit post-synthesis functional	467 checks passed per design

Fewer cycles at a higher hardware cost

Vivado 2023.2 · Artix-7 XC7A100T · 32-bit operands · Identical 10 ns constraints

Metric	Baseline	Booth	Change
Signed multiplication cycles	64	32	−50%
Slice LUTs	487	650	+33.5%
Flip-flops	266	365	+37.2%
DSP blocks	0	0	Unchanged
Inferred latches	0	0	None

Post-synthesis metric	Baseline	Booth
Worst setup slack	+0.442 ns	+0.496 ns
Worst hold slack	−1.079 ns	−1.117 ns

Tradeoff: 50% fewer signed multiplication cycles, using 33.5% more LUTs and 37.2% more flip-flops.